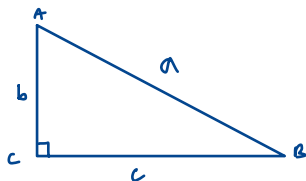


Pythagoras theorem & trigonometry

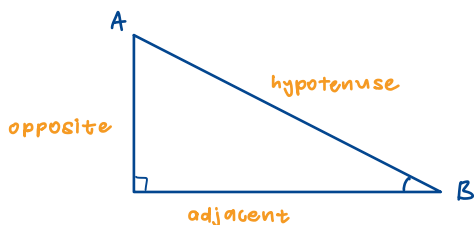
Pythagoras' Theorem



★ must be a right-angled Δ .

$$a^2 = b^2 + c^2 \text{ (Pythagoras' Theorem)}$$

Trigonometric Ratios of Acute Angles



★ must be a right-angled Δ .

TOA

$$\tan \theta = \frac{O}{A}$$

CAH

$$\cos \theta = \frac{A}{H}$$

SOH

$$\sin \theta = \frac{O}{H}$$

Obtuse Angle

$$\downarrow$$
$$90^\circ < \theta < 180^\circ$$

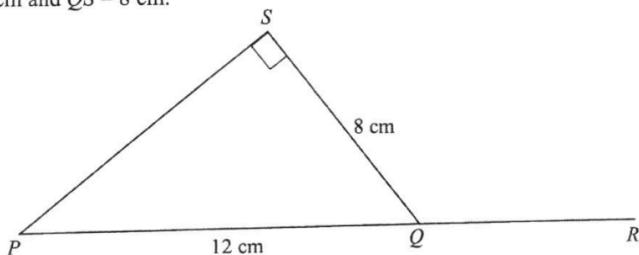
$$\sin \theta = \sin (180^\circ - \theta)$$

$$\cos \theta = -\cos (180^\circ - \theta)$$

$$\tan \theta = -\tan (180^\circ - \theta)$$



In the figure below, PQR is a straight line and PQS is a right-angled triangle.
 $PQ = 12$ cm and $QS = 8$ cm.



Giving your answer as a fraction in its lowest form, find the value of

(a) $\sin \angle SPQ$

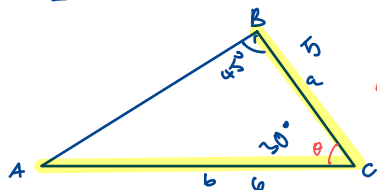
Answer _____ [1]

(b) $\cos \angle SQR$

Answer _____ [1]

Area of $\triangle$

$$\hookrightarrow \frac{1}{2}ab\sin C$$



← θ used is in between the 2 sides used.

Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The Sine Rule can be used to solve a triangle if the following are given:

- two angles and the length of one side; or
- the lengths of two sides and one non-included angle.

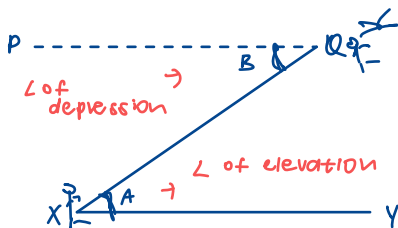
Cosine Rule

$$a^2 = b^2 + c^2 - 2bc \cos A$$

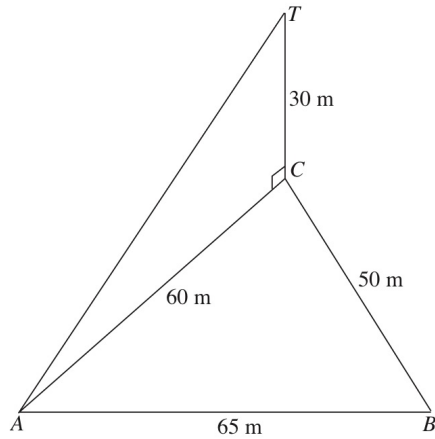
The Cosine Rule can be used to solve a triangle if the following are given:

- the lengths of all three sides; or
- the lengths of two sides and an included angle.

Angles of Elevation & Depression



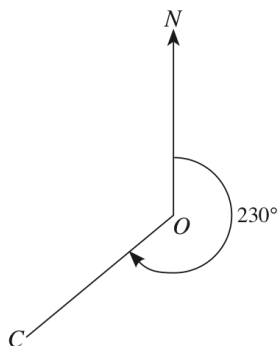
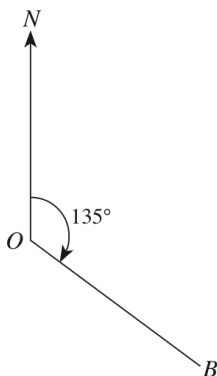
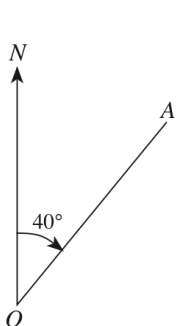
In the figure, A , B and C lie on level ground such that $AB = 65$ m, $BC = 50$ m and $AC = 60$ m. T is vertically above C such that $TC = 30$ m.



Find

- (i) $\hat{ACB}$,
- (ii) the angle of elevation of T from A .

The bearing of a point A from another point O is an angle measured from the north, at O , in a clockwise direction and is written as a three-digit number.
e.g.

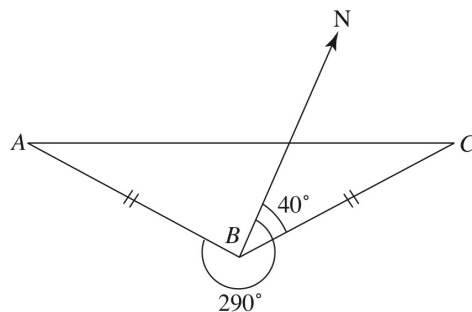


The bearing of A from O is 040 $^\circ$.

The bearing of B from O is 135° .

The bearing of C from O is 230° .

The bearing of A from B is 290° . The bearing of C from B is 040° . $AB = BC$.



Find

(i) $\hat{BCA}$,

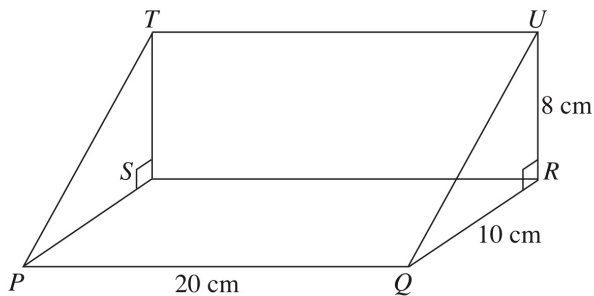
(ii) the bearing of A from C .

The diagram shows a right-angled triangular prism.

Find

(a) $\hat{SPT}$,

(b) PU .



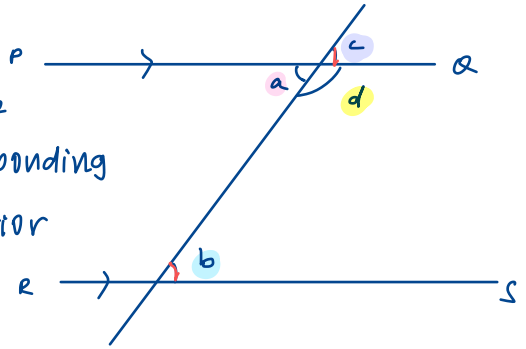
Geometrical properties of $\angle$ s.

If $PQ \parallel RS$

$\angle \angle a = \angle b$ (alt. $\angle$ s) $\rightarrow$ alternate

$\angle \angle c = \angle b$ (corr. $\angle$ s) $\rightarrow$ corresponding

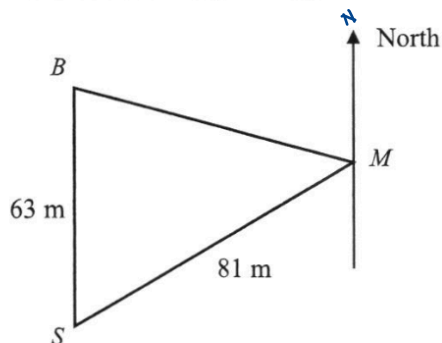
$\angle \angle b + \angle d = 180^\circ$ (int. $\angle$ s) $\rightarrow$ interior



When presenting answer,

(alt. $\angle$ s, $PQ \parallel RS$)

The diagram shows three points, B , S and M , on the level ground, which represent the bus-stop, the school and the MRT station respectively.



S is due south of B . The bearing of S from M is 245° . $SB = 63$ m and $MS = 81$ m

(a) Find the distance of BM .

Answer:m [3]

- (b) David has to walk to the library, L , from S . The bearing of L from S is 094° and $ML = 45$ m.
Given that there are two possible locations for L , find the two possible values of $\angle SLM$.

Answer:or..... [3]